place. The NSA may have paid for the network, but sysadmins like myself were the ones who really owned it.

The Read phase would involve dancing through the digital grid of tripwires laid across the routes connecting the NSA to every other intelligence agency, domestic and foreign. (Among these was the NSA's UK partner, the Government Communications Headquarters, or GCHQ, which was setting up dragnets like OPTICNERVE, a program that saved a snapshot every five minutes from the cameras of people video-chatting on platforms like Yahoo Messenger, and PHOTONTORPEDO, which grabbed the IP addresses of MSN Messenger users.) By using Heartbeat to bring in the documents I wanted, I could turn "bulk collection" against those who'd turned it against the public, effectively Frankensteining the IC. The agency's security tools kept track of who read what, but it didn't matter: anyone who bothered to check their logs was used to seeing Heartbeat by now. It would sound no alarms. It was the perfect cover.

But while Heartbeat would work as a way of collecting the files—far too many files—it only brought them to the server in Hawaii, a server that kept logs even I couldn't get around. I needed a way to work with the files, search them, and discard the irrelevant and uninteresting, along with those containing legitimate secrets that I wouldn't be giving to journalists. At this point, still in my Read phase, the hazards were manifold, due mainly to the fact that the protocols I was up against were no longer geared to monitoring but to prevention. If I ran my searches on the Heartbeat server, it would light a massive electronic sign blinking ARREST ME.

I thought about this for a while. I couldn't just copy the files directly from the Heartbeat server onto a personal storage device and waltz out of the Tunnel without being caught. What I could do, though, was bring the files closer, directing them to an intermediate way station.

I couldn't send them to one of our regular computers, because by 2012 all of the Tunnel had been upgraded to new "thin client" machines: small helpless computers with crippled drives and CPUs that couldn't store or process data on their own, but did all of their storage and processing on the cloud. In a forgotten corner of the office, however, there was a pyramid of disused desktop computers—old, moldering legacy machines the agency had wiped clean and discarded. When I say old here, I mean young by the standards of anyone who doesn't live on a budget the size of the NSA's. They were Dell PCs from as recently as 2009 or 2010, large gray rectangles of comforting weight, which could store and process data on their own without being connected to the cloud. What I liked